Yucheng (Cecilia) Zhang

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Education

Northwestern University, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL

The University of Texas at Austin, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX

Relevant Coursework: Stochastic Processes, Probability and Statistics, Real Analysis, Data Structures, Data Analytics, Database Systems, Artificial Intelligence

Technical Skills

Quantitative: Probability, statistics, predictive modeling, credit risk, feature engineering, model validation, backtesting, ROC/AUC
Programming: Python, SQL, R, MATLAB, SAS, Git, Excel; Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib
Data Systems: SQL Server, Oracle, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI
Financial Analytics: Risk segmentation, portfolio monitoring, fraud analytics, transaction monitoring, SEC filing analysis

Quantitative Analytics and Financial Data Experience

Enterprise Risk Analyst Intern — May 2025 – Dec. 2025
Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI — Dallas, TX

- Developed an end-to-end credit risk modeling pipeline integrating PayNet inquiries and internal loan servicing data through SQL Server ETL, feature engineering, Python modeling, validation, and reporting automation.
- Engineered borrower, contract, and asset-level features including loan-to-value ratio, account tenure, inquiry patterns, vehicle segments, and servicing attributes for truck, trailer, and used-vehicle portfolios.
- Validated Logistic Regression and Random Forest outputs with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing scorecard segmentation updates.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by **10%**.

Financial Data Development Engineer – Fraud Detection — Jul. 2024 – Aug. 2024
Caitong Securities Ltd | SQL, Oracle, Stored Procedures — Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by **5%**.
- Implemented parameterized fraud-detection logic and customer risk scoring from transactional and behavioral streams, reducing false positives by **2%** in daily monitoring workflows.
- Built reviewer-facing risk logic for fraud analytics and AML-related workflows, translating transaction patterns into prioritized risk signals for operational review.

FinChat – AI-Powered Financial Analysis System — Jan. 2023 – May 2024
Southern Methodist University | Python, FinGPT, LangChain, CrewAI — Dallas, TX (Remote Hybrid)

- Built ETL workflows to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial reports for metric extraction, trend detection, and company-level comparison.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving filing-analysis accuracy by **3%** and reducing manual review time.
- Connected filing preprocessing, retrieval, metric extraction, and financial-question workflows into a reusable prototype for comparing companies across reporting periods.

Quantitative Research and Technical Projects

CodeAssist – AI Feedback System | GitHub — Aug. 2025 – Present
University of Texas at Austin | Flask, React, PostgreSQL, OpenAI — Austin, TX

- Built a full-stack AI feedback system with context filtering, usage limits, model-provider settings, and instructor controls, connecting software engineering with applied AI tooling.
- Developed Flask REST APIs and PostgreSQL models for AI feedback settings, student submissions, request limits, cooldown policies, JSON serialization, and role-based access.
- Implemented context permissions so assignment descriptions, code, uploaded files, tests, execution results, and outputs are shared with AI providers only when instructor policies allow them.

Researcher & TA – Biological Data Analytics — Jan. 2024 – Dec. 2024
University of Texas at Austin | R, Linux, AWS, Docker — Austin, TX

- Processed **250GB** of RNA-seq data with reproducible R/Linux workflows; performed Voom/Limma differential-expression analysis, PCA, and permutation-based FDR to identify 48 key genes.
- Applied Salmon, RSEM, Trinity, and samtools for transcript quantification, achieving a **98.7% mapping rate** and training incoming students on reproducible analysis pipelines.
- Automated normalization, PCA, and volcano-plot workflows in R/ggplot2, improving interpretation efficiency by 15% for time-series biological datasets.

Machine Learning Research Assistant – Neuroengineering Lab — Jan. 2023 – May 2024
Python, MATLAB, MNE, RF, SVM, KNN, XGBoost — Remote

- Processed **35GB** of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, statistical testing, and emotion-recognition model evaluation.
- Applied statistical tests and feature-extraction methods to evaluate cognitive impairment patterns and emotion-recognition signals across experimental conditions.
- Evaluated RF, SVM, KNN, and XGBoost models, contributing analysis used in peer-reviewed work in *Frontiers in Neurology*, *Frontiers in Neuroscience*, and IEEE ICIPCA.

Selected Publications

- Coauthored **5 peer-reviewed publications** applying machine learning and statistical analysis to computer vision, neuroengineering, and social-media data.
- Selected venues include *Expert Systems with Applications*, *Frontiers in Neurology*, *Frontiers in Neuroscience*, and IEEE ICIPCA.